Machine Learning & Artificial Intelligence for Data Scientists: Welcome to the course

Ali Gooya
ali.gooya@glasgow.ac.uk
School of Computing Science

Who are we?



Prof Fani Deligianni

Machine Learning and Healthcare



Dr Ali Gooya
Senior Lecturer

Machine Learning and Medical Imaging

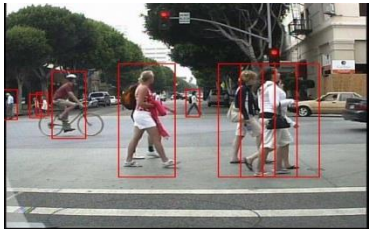
Examples of applied ML



Spam filtering



Face detection



Pedestrian detection



Movie recommendation



Voice recognition

What is Machine Learning?

Machine Learning (ML):

The field of study that gives computers the ability to learn without being explicitly programmed.

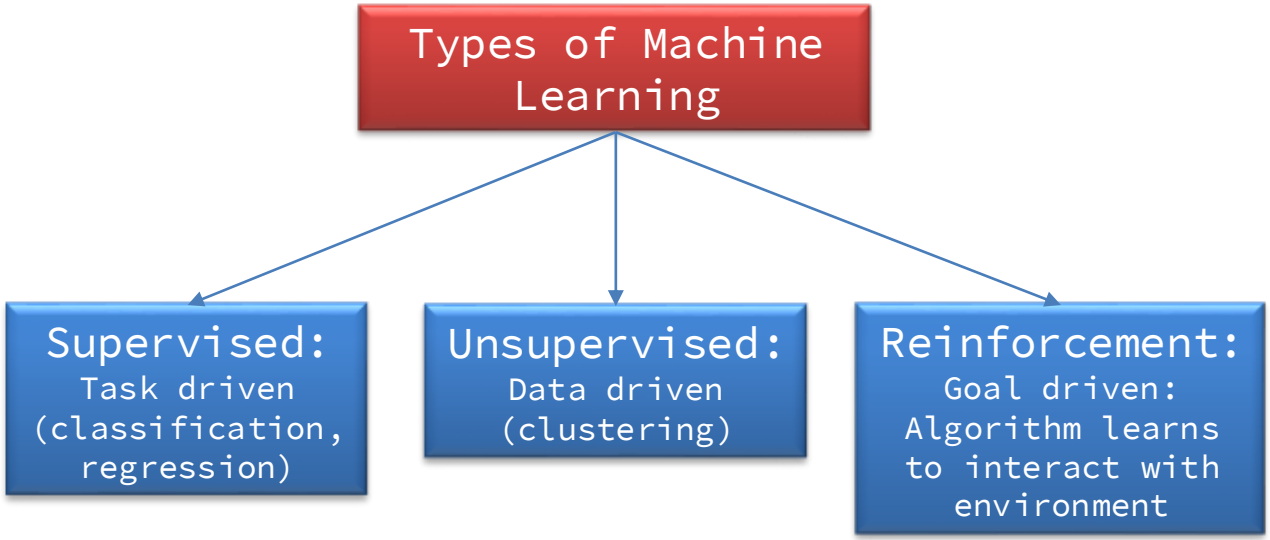
Arthur Lee Samuel was an American pioneer in the field of computer gaming and artificial intelligence. He popularized the term "machine learning" in 1959.



What is Machine Learning?

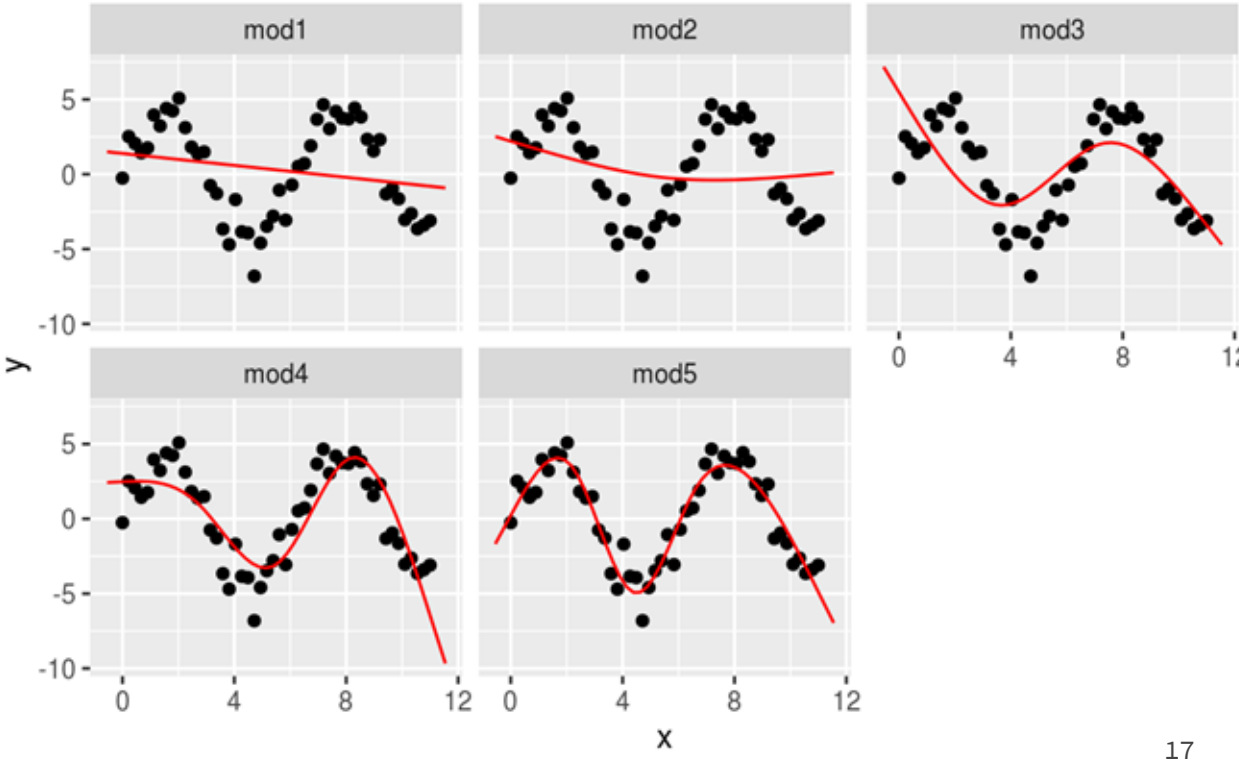
- Machine learning starts with **data**.
- **Data** is derived from **objects**:
  - Observations of people (preferences, health, etc)
  - Observations of the world (images, sounds, etc)

Types of Machine Learning



Supervised Learning: Regression

- Learning a continuous function from a set of examples.
- Example: Predicting stock prices (x might be time or some other variable of interest).



What do we want to achieve?

“All models are wrong, but some are useful.”

George E. P. Box

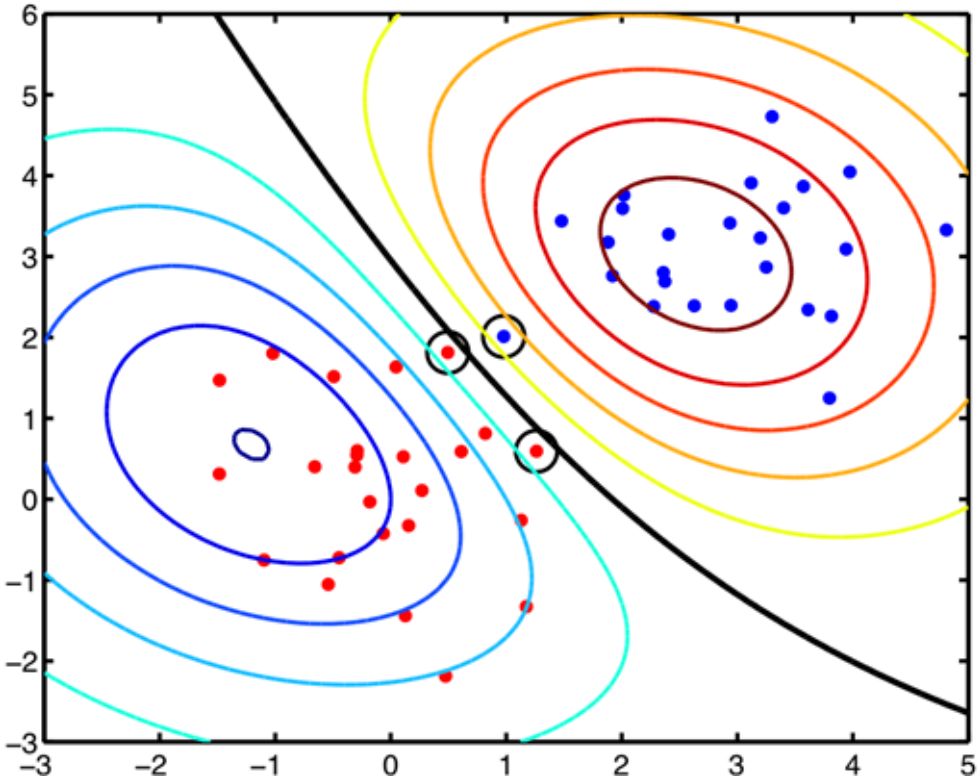


Now it would be very remarkable if any system existing in the real world could be exactly represented by any simple model. However, cunningly chosen parsimonious models often do provide remarkably useful approximations. For example, the law PV = RT relating pressure P, volume V and temperature T of an “ideal” gas via a constant R is not exactly true for any real gas, but it frequently provides a useful approximation and furthermore its structure is informative since it springs from a physical view of the behavior of gas molecules.

For such a model there is no need to ask the question “Is the model true?”. If “truth” is to be the “whole truth” the answer must be “No”. The only question of interest is “Is the model illuminating and useful?”.

Supervised Learning: Classification

- Learning a rule that can separate objects of different types from one another
- Examples: Disease diagnosis, spam email detection.

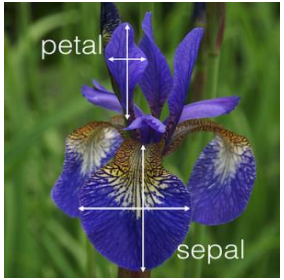


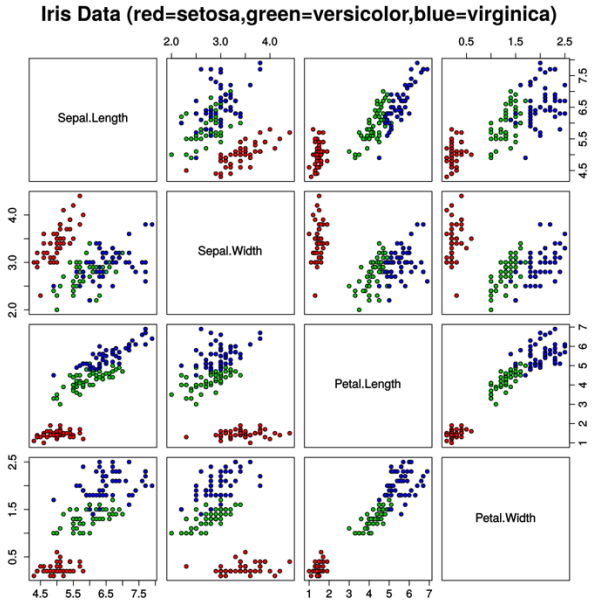
Machine Learning Process

Data Collection:
For example, many images of various iris flowers



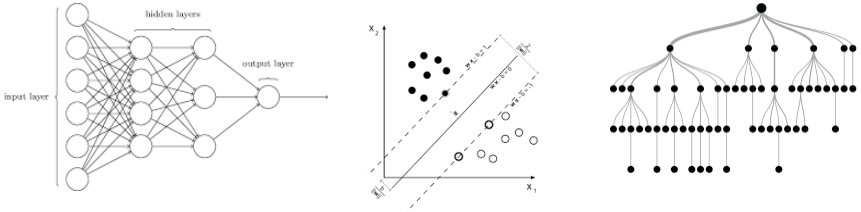
Feature Selection:
E.g., length/widths of petal and sepal in each flower



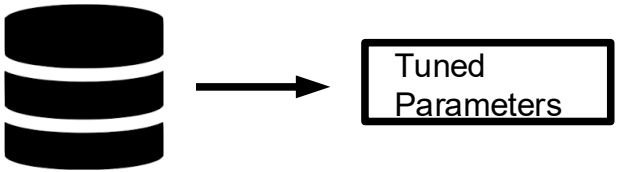


Machine Learning Process

Model choice



Training



Data

Evaluation

		PredictedClass		
		Setosa	Versicolor	Virginica
Actual Class	Setosa	**14**	1	1
	Versicolor	1	**11**	3
	Verginica	1	3	**10**

Machine Learning Process

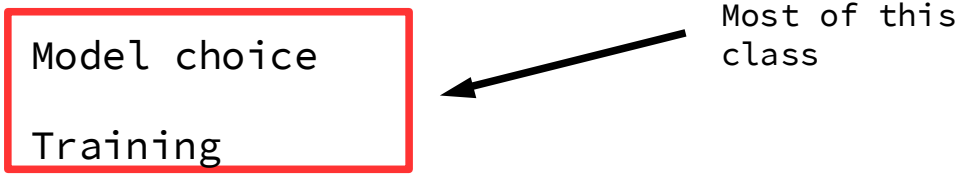
Data Collection

Feature Selection

Model choice

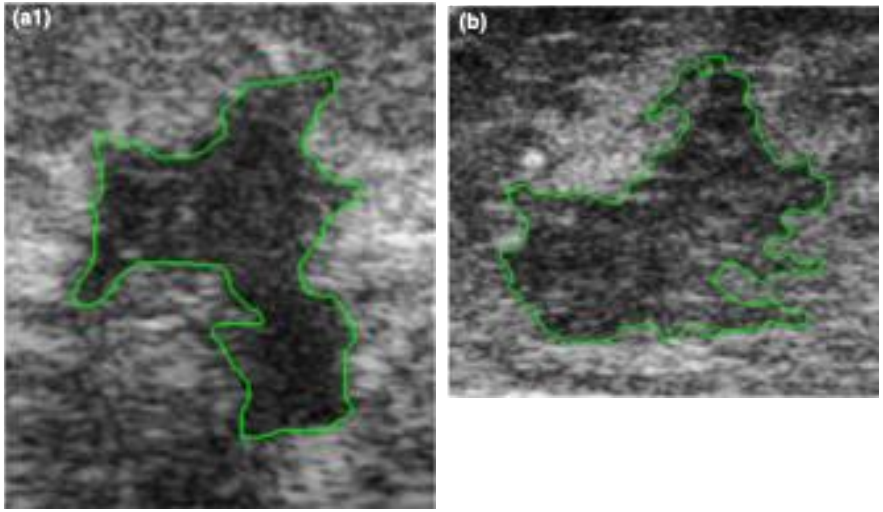
Training

Evaluation

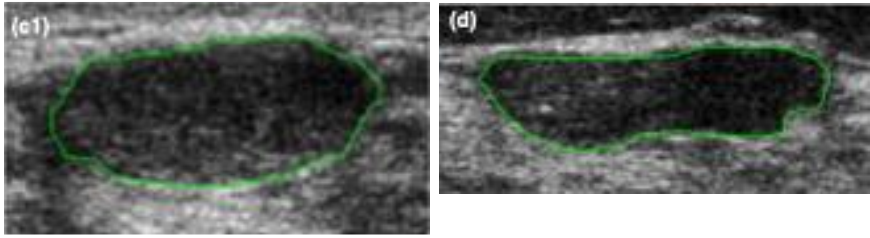


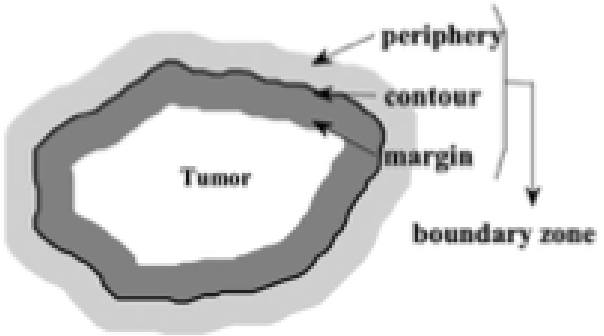
Classification of breast lesions in ultrasonography using sparse logistic regression and morphology-based texture features, Medical Physics, 2019

Samples of malignant tumours



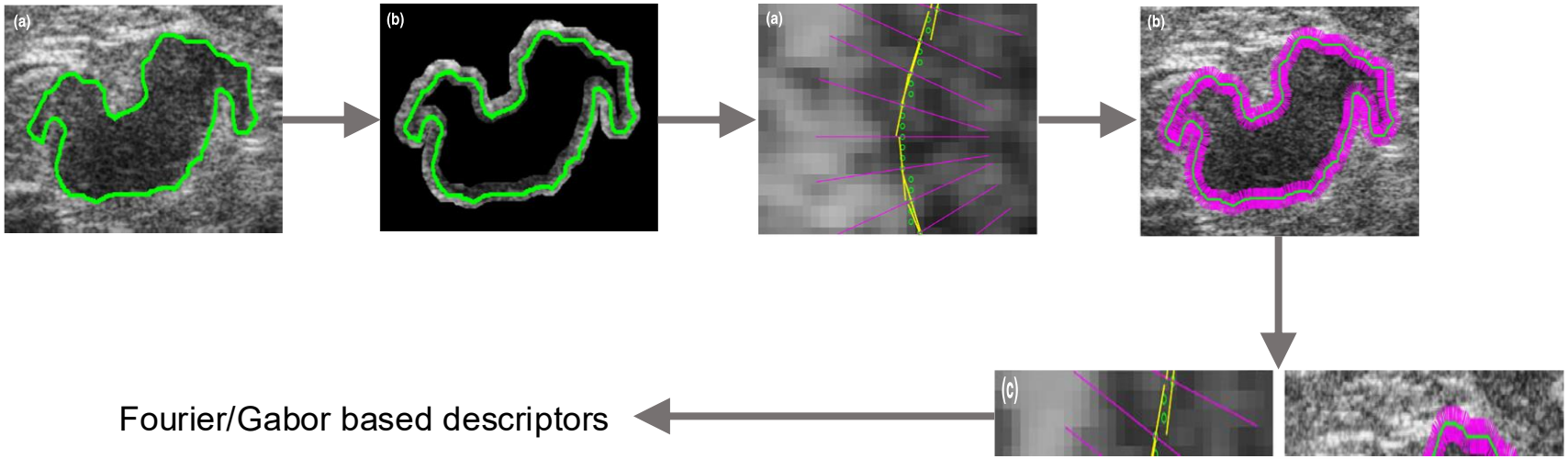
Samples of benign tumours



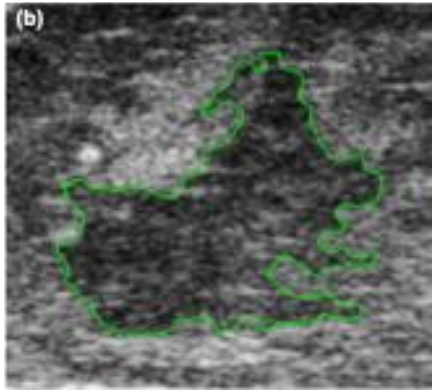


Classification of breast lesions in ultrasonography using sparse logistic regression and morphology-based texture features, Medical Physics, 2019

Extracting texture-based features from tumour boundary

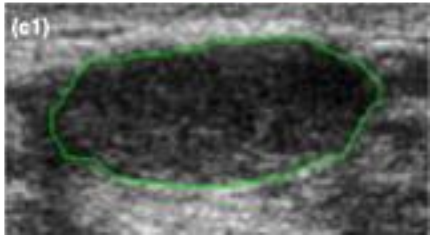


Classification of breast lesions in ultrasonography using sparse logistic regression and morphology-based texture features, Medical Physics, 2019



Malignant



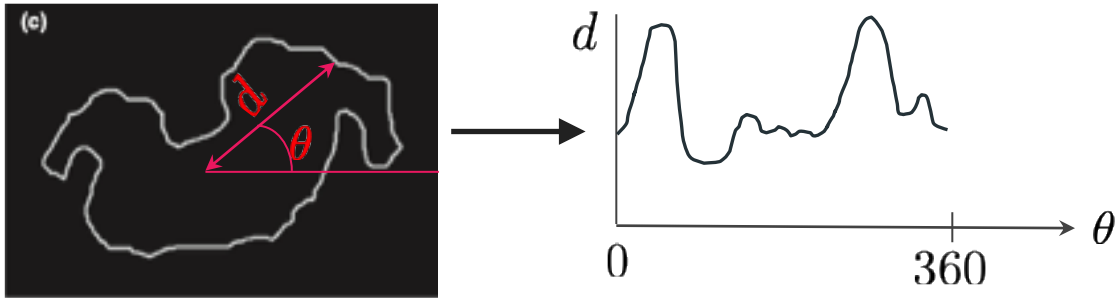


Benign



Classification of breast lesions in ultrasonography using sparse logistic regression and morphology-based texture features, Medical Physics, 2019

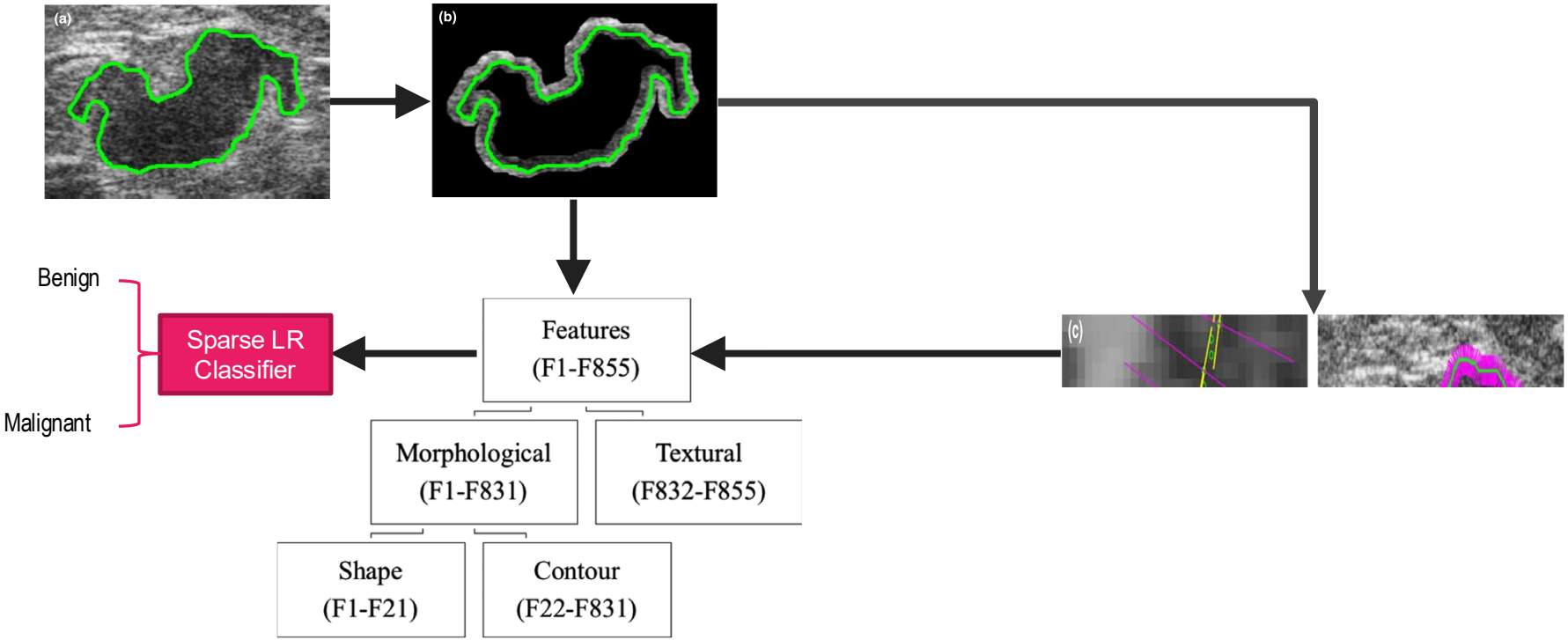
Extracting contour-based features



Fourier descriptors

Classification of breast lesions in ultrasonography using sparse logistic regression and morphology-based texture features, Medical Physics, 2019

- Applied to 104 breast US images of tumours, including 72 benign and 32 malignant

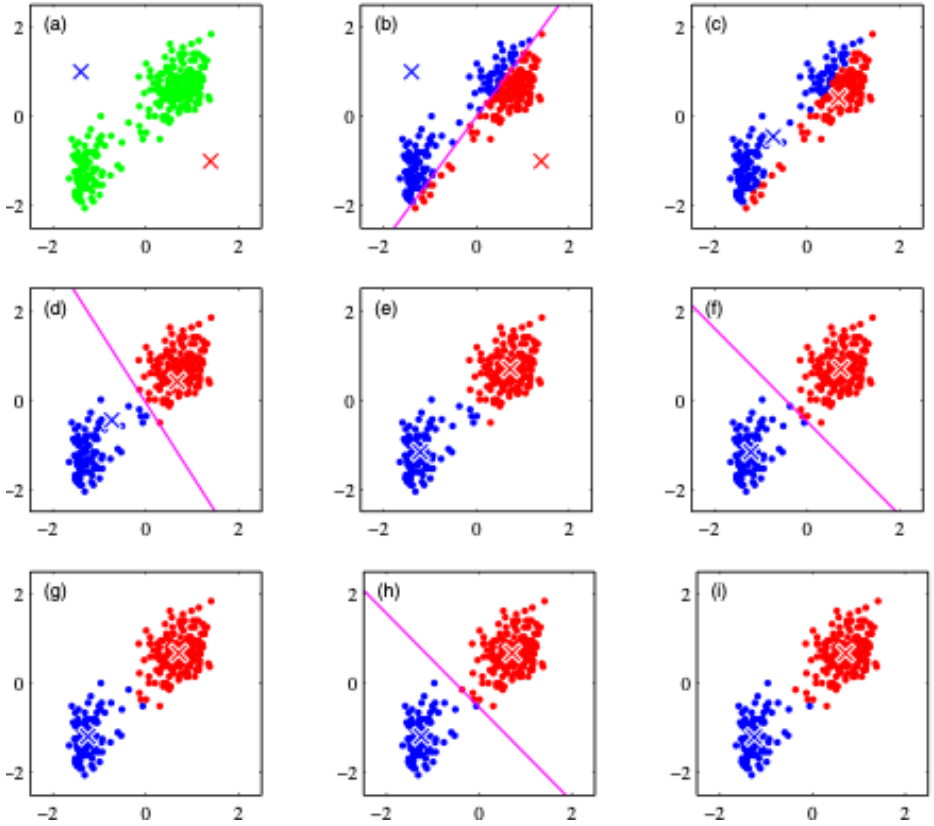


Unsupervised Learning: Clustering

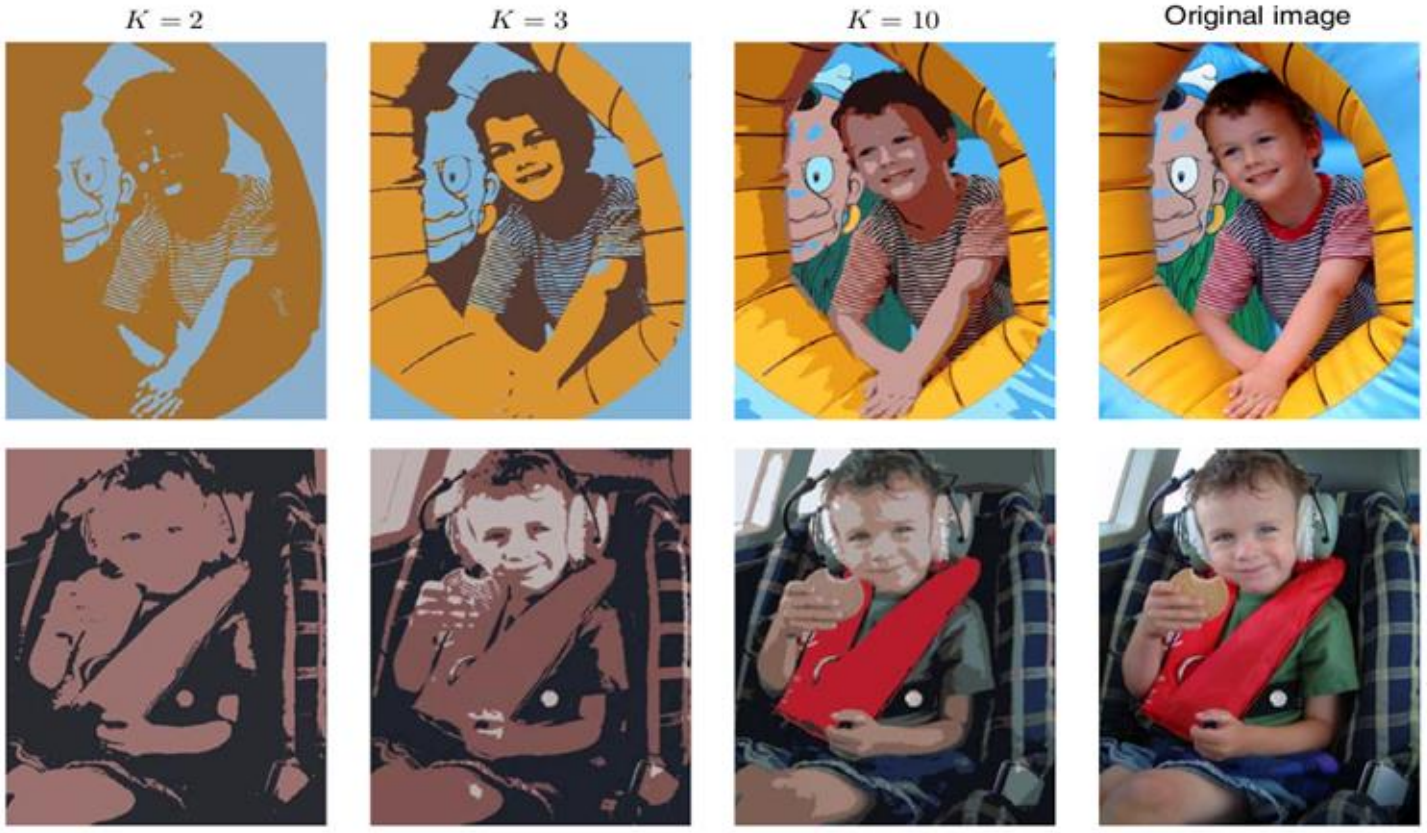
- Finding groups of similar data points.
- Example: clustering re-scaled Old Faithful dataset



Observation	Eruption Duration (min)	Waiting Time (min)
1	3.6	79
2	1.8	54
3	3.333	74
4	2.283	62
5	4.533	85
6	2.883	55
7	4.7	88
8	3.6	85
9	1.95	51
10	4.35	85
11	1.833	54
12	3.917	84
13	4.2	78
14	1.75	47

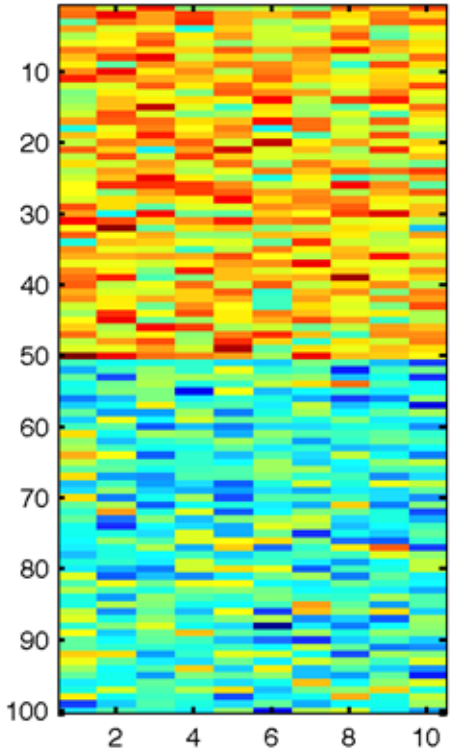


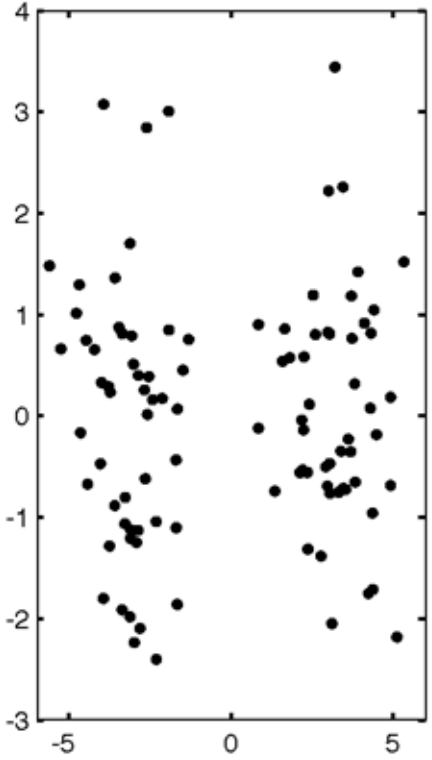
Clustering (color quantisation)



Projection (Dimensionality Reduction)

- Reducing the number of variables - e.g. from 10 to 2.
- Visualising complex data.

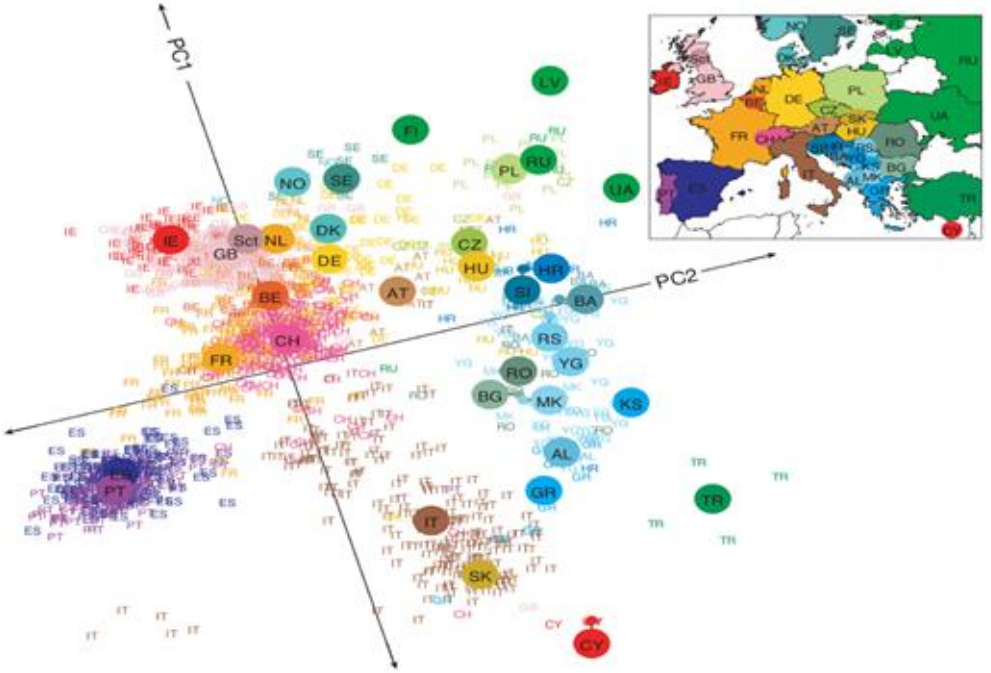




Genes mirror geography within Europe

Novembre et al. (2008) doi:10.1038/nature07331

- PCA was applied to produce a two-dimensional visual summary of the observed genetic variation in Europe.
- The resulting figure bears a notable resemblance to a geographic map of Europe.
- Genetic coordinates are used to predict subjects' locations.



Course structure

- Topics covered
  - What is “learning from data”
  - Introduction to the problems of Regression, Classification, Clustering, and Projection.
  - Training, validation, and testing.
  - Performance metrics and the importance of robust baselines.
  - Common pitfalls.
  - Presentation of results.

Course structure

- **2 units of case studies** covering different datasets, visualisation techniques, models and learning algorithms. Taught by multiple staff.
- **Feedback from last year**: use cases (and grouping) will be released much earlier:

Coursework 1	20.00%	Friday 30 Oct 2026	Friday 14 November 2026
Coursework 2	20.00%	Friday 6 Nov 2026	Friday 20 November 2026

Case study

- A dataset and predictive / exploratory problem to be solved.
- An introduction to one or more algorithms.
- A practical session (1 hr in lab plus 1hr in students' owen time).
- A wrap-up session .

Assessment and feedback

- Exam: 60%, topics covered in lectures.
- Coursework: 40%, 2 case studies.
- Written feedback for the coursework.
- 1-1 help from GTAs and demonstrators in the labs.
- Lectures where solutions or common issues are discussed.

Don’t panic! Have fun!